AP PRECALCULUS

Unit 1 • Lesson 1.4 • Guided notes

# Topic 1.4 • Polynomial Functions and Rates of Change

Guided notes • Fill in the blanks and work the examples in the boxes

**Learning goals**

- Identify the degree, leading term and leading coefficient of a polynomial in any form.
- Locate local and global extrema from a graph, including on closed intervals.
- Identify points of inflection and describe the rate of change between turning points.

**Vocabulary.** polynomial, degree, leading term, leading coefficient, local (relative) extremum, global (absolute) extremum, point of inflection.

## 1. Polynomial functions

A polynomial of degree $n$ is $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ with $a_n$ _____. $a_n x^n$ is the _____ and $a_n$ the _____. The degree of a product is the _____ of the degrees of the factors.

**Example 1**

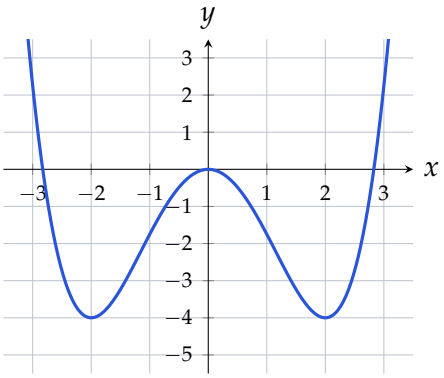
Give the degree and leading coefficient of $p(x) = -2x(x-1)^2(x+3)^3$.

## 2. Extrema

**Key idea**

- A **local maximum** occurs where $p$ changes from _____.
- A **local minimum** occurs where $p$ changes from _____.
- The **global maximum** on an interval is the _____ output value on that interval; on a closed interval it can occur at an _____.
- A polynomial of degree $n$ has at most _____ local extrema.

**Example 2 • Reading extrema**



The graph of $p(x) = \frac{1}{4}x^4 - 2x^2$ is shown. Find all local extrema, the global extrema on $(-\infty, \infty)$, and the global maximum on $[0, 3]$.

## 3. Points of inflection

**Key idea**

A point of inflection is where the graph changes _____: the rate of change switches from increasing to decreasing, or the reverse. Between a local maximum and the next local minimum there is at least one point of inflection.

**Example 3**

$p(x) = x^3 - 6x^2 + 9x$ has a local maximum at $(1, 4)$, a local minimum at $(3, 0)$ and a point of inflection at $(2, 2)$. Describe $p$ and its rate of change.

**Common mistakes**

- Giving the $x$-value when asked for the extreme *value*, or the reverse.
- Forgetting the endpoints when looking for global extrema on a closed interval.

**Check your understanding**

1. A degree-6 polynomial has at most ____ turning points.
2. A cubic with no local extrema: _____.